



Hindustan Petroleum Corporation Limited
Corrosion Research Laboratory
HP Green R&D Centre
Bengaluru-560 067

MATERIAL SAFETY DATA SHEET

Additive for mitigating Ammonium Chloride Corrosion in Hydrotreaters (HP-CorrInAc)

SECTION 1. CHEMICAL IDENTIFICATION

Name : Corrosion Inhibitor for Ammonium Chloride Corrosion
Application : Inhibiting corrosion in Ammonium Chloride environment in Hydrotreater Units
Use : Corrosion Inhibitor
Address : Hindustan Petroleum Green R&D Centre,
KIADB industrial area, Tarabahalli village,
Devanagundi, Hoskote Taluk,
Bangalore-560067.

SECTION 2. COMPOSITION

Name	Content
Amine	10-20 wt%
Heterocyclic ether	10-20 wt%
Anti-oxidant	10-20 wt%
Solvent	50-80 vol%

SECTION 3. PHYSICAL AND CHEMICAL PROPERTIES

Appearance : Clear & Brown colored liquid
Density : 1.084 g/cc
pH : 12
Viscosity : 4.285 mm²/s

National Fire Protection Association (U.S.A.):

Health : 3



Flammability : 1
Reactivity : 0
Specific hazard : N/A

SECTION 4. HAZARDS IDENTIFICATION

Most important hazards: Harmful if swallowed and inhaled.
Causes severe skin burns and eye damage
May cause respiratory irritation

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May cause damage to organs through prolonged or repeated exposure
Toxic in contact with skin or if inhaled

SECTION 5. FIRST-AID MEASURES

Ingestion	Wash out mouth with water. Call a physician. Do NOT induce vomiting.
Eye contact	Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
Skin contact	Wash contaminated clothing before reuse IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower.
Inhalation	IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing.
Notes to physician	In all cases get medical attention.

SECTION 6. FIRE FIGHTING MEASURES

Extinguishing Media	Water spray, carbon dioxide (CO ₂), dry chemical, alcohol-resistant foam. Water mist maybe used to cool closed containers.
Firefighting instruction	Cool containing vessel with water to prevent pressure build up. Respiratory, skin & eye protection required
Special hazards	Thermal decomposition can lead to release of irritating gases and vapors. The product causes of eyes burn, skin and mucous membranes. Vapors may form explosive mixtures with air. In the event of fire and/or explosion do not breathe fumes

SECTION 7. ACCIDENTAL RELEASE MEASURES

Personal Precautions	Use personal protective equipment as required. Ensure adequate ventilation. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges.
Environmental Precautions	Should not be released into the environment. See Section 12 for additional Ecological Information. Do not flush into surface water or sanitary sewer system. Prevent further leakage or spillage if safe to do so. Prevent product from entering drains



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Methods for Containment and Clean up	Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment
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SECTION 8. HANDLING AND STORAGE

Handling	Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed, then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges. Do not taste or swallow.
Storage	Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. Keep away from heat, sparks and flame. Keep in properly labeled containers. Flammables area. Incompatible Materials. Strong oxidizing agents.

SECTION 9. EXPOSURE CONTROLS AND PERSONAL PROTECTION

Exposure Limit	Avoid prolonged or repeated exposure
Engineering measure	Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment. Ensure adequate ventilation, especially in confined areas.
Personal protection	Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910. Impervious clothing. Chemical resistant apron. Boots. Impervious gloves.
TWA	20 ppm
STEL	30 ppm

SECTION 10. STABILITY AND REACTIVITY

Conditions to avoid	Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition.
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	Exposure to moist air or water. Exposure to air or moisture over prolonged periods
Incompatibility	Strong oxidizing agents
Decomposition Products	Carbon monoxide (CO), Carbon dioxide (CO ₂), Nitrogen oxides (NO _x), Thermal decomposition can lead to release of irritating gases and vapor
Hazardous polymerization	Does not occur

SECTION 11. TOXICOLOGICAL INFORMATION

Toxicity Data	Hazardous in case of skin contact (irritant)
Route of entry	Absorbed through skin. Eye contact
Effects of overexposure	Causes skin irritation and burn. Harmful if swallowed. Causes gastrointestinal / digestive track irritation. Causes eye irritation and burns
Medical Conditions Generally Aggravated by exposure	Harmful effects on skin lungs and eyes in case of prolonged exposure

SECTION 12. ECOLOGICAL INFORMATION

- Do not empty into drains.
- Soluble in water Persistence is unlikely based on information available

SECTION 13. DISPOSAL CONSIDERATIONS

- Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

SECTION 14. TRANSPORT INFORMATION

UN hazard classification : Non-flammable
Hazards class : 8
Class : I?

SECTION 15. REGULATORY INFORMATION

- Harmful by inhalation, in contact with skin and if swallowed. If you feel unwell, seek medical advice immediately.
 - In case of contact with eyes, rinse immediately with plenty of water and seek medical advice
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- Take off immediately all contaminated clothing
- Wear suitable protective clothing, gloves and eye/face shield

SECTION 16. OTHER INFORMATION

The information and recommendations in literature and on labels are to the best of our knowledge reliable. However, nothing herein is to be construed as a warranty or representation. Users should make their own tests to determine the applicability of such information or the suitability of any products for their own particular purpose. The license is granted to make unlimited paper copies for internal use only.